Stelios Zacharioudakis

MACHINE LEARNING ENGINEER | APPLIED AI & ML SYSTEMS

New York City Metropolitan Area | +1 (332) 203-1386 | stelios@stelioszach.com
stelioszach.com | github.com/stelioszach03 | linkedin.com/in/stelios-zach

Machine learning engineer with end-to-end ownership of a deployed association platform. Build multilingual LLM workflows, evaluate coding agents against baselines and benchmark GPU inference.

PROFESSIONAL EXPERIENCE

Paphos Medical Association - Head Engineer (Pro Bono) | Jun 2022 - Jul 2026

- Owned architecture, implementation and Linux deployment for AsklepiosMed, a production platform I built and donated to consolidate member services and association administration.
- Built the React, Express and PostgreSQL stack, including role-based access, member workflows and administrative publishing.

SELECTED ENGINEERING PROJECTS

AsklepiosMed - Multilingual NLP workflows

- Extended the platform after the pro bono role with LLM-assisted drafting and proofreading for administrative publishing.
- Built Greek-to-English, Russian and Simplified Chinese translation with terminology checks, structured output validation, fallback handling and editor review.

ForgeRL / ForgeBench - Reinforcement learning and agent evaluation

- Built a sandboxed coding-agent benchmark with fitted-Q routing and persistent spend controls. Compared six policies in 162 transfer episodes and ran a separate 1,296-attempt follow-up.
- On the follow-up primary slice, strong-only routing succeeded on 131 of 144 attempts at lower accounted model cost than fitted-Q routing, which succeeded on 128 of 144.

Inference Systems Lab - GPU serving evaluation

- Benchmarked 16,384 vLLM requests across RTX 4090/A100, two models and four repeats; measured 1.23-1.47x cached/uncached throughput ratios for shared-prefix workloads at concurrency 32, with output mismatches disclosed.

MTA-Scan - Streaming data and temporal evaluation

- Built a monitored GTFS-Realtime pipeline and FastAPI/Mapbox workspace with freshness checks and forecast expiry. Verified 44 daily exports and evaluated arrival-spacing proxy forecasts over 28 days.

MRI reconstruction - PyTorch and generative models

- Reimplemented a published score-based reconstruction method with Flax-to-PyTorch checkpoint conversion; reproduced reference PSNR within 0.15 dB on 256 BraTS slices at 4x, 8x and 24x undersampling.
- Evaluated reconstruction quality, uncertainty and retrospective domain shift; published reproducible evidence and independent CPU ensemble diagnostics.

TECHNICAL SKILLS

ML & AI: Python, PyTorch, reinforcement learning, LLM agents, world models, generative models, NLP, vLLM, NumPy, SciPy, scikit-learn and experimental evaluation.

Engineering: FastAPI, React, Node.js/Express, JavaScript, PostgreSQL, Redis, Linux, Nginx, Docker, Git and automated testing.

EDUCATION

BSc in Computer Science | National and Kapodistrian University of Athens, Jun 2026
Degree: Very Good | Thesis: Excellent (10/10), deposited September 2026.